Nokia 1830 PSS iROADM cards

As part of the 1830 Photonic Service Switch (PSS) product family, iROADM cards enable efficient, cost-effective metro, regional and long-haul core optical networks. By integrating switched gain amplifiers and wavelength selective switching (WSS) into a single card, iROADM cards reduce the cost and physical size of deploying modern optical networks.

The new generation of integrated ROADM (iROADM) cards simplifies the deployment of advanced optical networks. Each iROADM card includes all the functions needed to support an optical degree, including the amplifiers, WSS, OCM, OSC, and OTDR monitor ports. Integrated switched gain amplifiers enable a single card type to be deployed over short to ultra-long reach spans. The reduced card size of the iROADM is ideal for applications where space is at a premium, such as co-location sites. The iROADM family consists of the IR4, IR9, IRDM20, and iRDM32. The difference between the cards is the number of available WSS ports and supported node size.



Benefits

- High-performance switched gain amplifiers support any span distance with a single card type
- Simplified engineering, ordering and sparing of WDM systems
- Integrated, compact size reduces overall space requirements
- Lower WDM optical networking costs
- Integrated OTDR ports
- Full 4800 GHz (96 channel) capacity

Applications

- Metro, regional, long-haul optical networking applications
- Carrier, ISP, web, government, enterprise, public networks
- Supports all optical networking architectures, terminal, classic mux/demux ROADM, C-F, and CDC-F ROADMs
- Mix-and-match 1830 PSS shelf size and capacities to each application

Product description

The Nokia 1830 iROADM units integrate all functions needed to support a ROADM optical degree per card, including a switched gain pre-amp, post-amp, WSS, optical channel monitor (OCM), and optical supervisory channel (OSC).

The integration of all functions into a single card reduces the size and cost of deploying optical networks. In addition, carriers no longer need to engineer, order, purchase and spare multiple cards.

Switched-gain amplifiers enable a single card type to be used over any span distance, including short, medium, long-haul (LH) or ultra-long-haul (ULH).

The amplifier settings adjust to the network application, eliminating the need for separate card types.

With central office, data center and co-location space at a premium, integration and small size make the iROADM ideal for any network application.

The iROADM family includes several versions, each supporting different node size and drop capacities.

Nokia supported products

The iROADM units are supported on Nokia 1830 PSS-8, -16ii, -32 and PSI-L platforms.

Figure 1. ROADM node examples

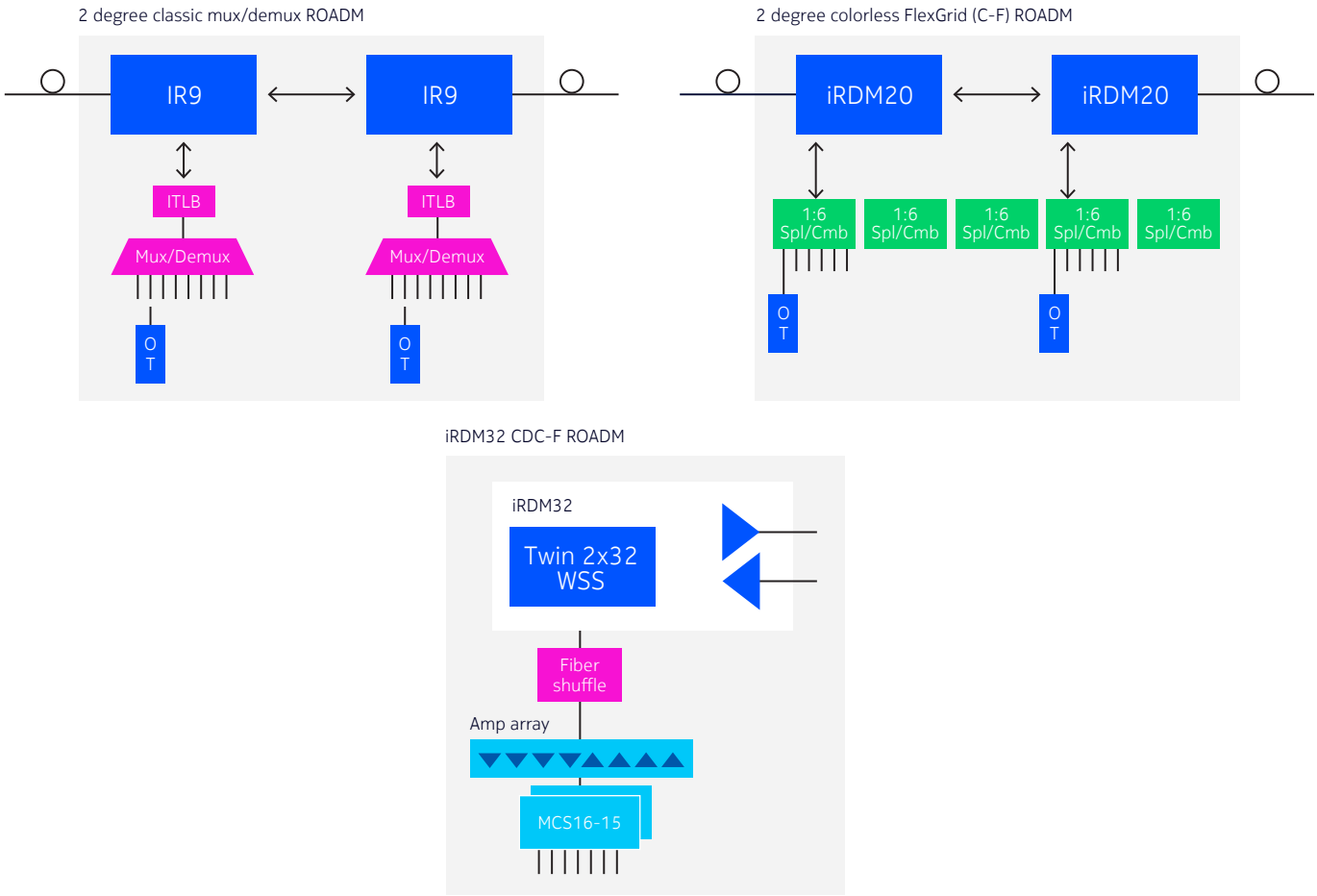


Table 1. iROADM versions

iROADM	Part #	Description
iR4	3KC71462AA	Integrated ROADM, 1x4 WSS, variable gain amp
iR9	3KC70869AA	Integrated ROADM, 1x9 WSS, sw gain amp
iR9LP	3KC70869AB	Integrated ROADM, 1x9 WSS, sw gain amp, limited power (20 dB)
iRDM20	3KC69751AA	Integrated ROADM, twin 2x20 WSS, sw gain amp
iRDM20LP	3KC69751AB	Integrated ROADM, twin 2x20 WSS, sw gain amp, limited power (20 dB)
iRDM32	3KC70163AA	Integrated ROADM, twin 2x32 WSS, sw gain amp
iRDM32LP	3KC71158AA	Integrated ROADM, twin 2x32 WSS, sw gain amp, limited power (20 dB)
iRDM32L	3KC70342AA	Integrated ROADM, twin 2x32 WSS, sw gain amp, L-Band

Table 2. Technical specifications

Specifications	IR4	IR9	iRDM20	iRDM32
Application	Metro, regional	Metro, retional, LH/ULH	Metro, regional, LH/ULH	Metro, regional, LH/ULH
Channels	4800 GHz	4800 GHz	4800 GHz	4800 GHz
WSS	1 x 4 Broadcast & Select	1 x 9 Broadcast & Select	Twin 2 x 20 Route & Select	Twin 2 x 32 Route & Select
Amplifier	Var gain pre-amp Post amp	Sw gain pre-amp Post amp	Switched gain pre-amp Post amp	Switched gain pre-amp Post amp
OTDR Ports	Integrated	Integrated	Integrated	Integrated
Power consumption	< 75 W	< 75 W	< 75 W	< 75 W
Operating environment	Normal 5°C to 40°C (41°F to 104°F) Humidity 5% to 85%			
Physical	1-slot, full height			
Shelf options	1830 PSS-8, PSS-16ii, PSS-32, PSI-4L, PSI-8L			

About Nokia

At Nokia, we create technology that helps the world act together.

As a B2B technology innovation leader, we are pioneering networks that sense, think and act by leveraging our work across mobile, fixed and cloud networks. In addition, we create value with intellectual property and long-term research, led by the award-winning Nokia Bell Labs.

Service providers, enterprises and partners worldwide trust Nokia to deliver secure, reliable and sustainable networks today – and work with us to create the digital services and applications of the future.

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Document code: (June) CID201270